Ricardo (Zhicong) Xie

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Simulation / Analysis Intern | FEA, CFD and Mechanical Design Checks

PROFILE

- Mechanical engineering master's student with project work in Abaqus structural analysis, ANSYS Fluent/CFX thermal-flow modelling and model credibility checks.
- I am strongest where the job needs assumptions, load cases, boundary conditions, mesh checks, result interpretation and honest limits.
- Project screenshots and source links: https://zxie588-alt.github.io

WORK RIGHTS AND AVAILABILITY

- NZ student visa. Eligible to work up to 25 hours/week during semester and full-time during scheduled university breaks.
- Available full-time from 17 Nov 2026 to late Feb 2027. Auckland preferred; open to NZ relocation for the right internship opportunity.

EDUCATION

University of Auckland - Master of Engineering Studies, Mechanical Engineering | 2026 - Present

Current study/projects in building services, smart manufacturing, CFD/thermal modelling, composite structures and engineering methods.

Huazhong Agricultural University - Bachelor of Engineering, Mechanical Engineering / Mechatronics | 2021 - 2025

Mechanical design, CAD, electrotechnics, sensors, embedded control and mechatronic system integration.

PROJECT WORK

Composite Formula SAE Front Wing Analysis

Abaqus/Standard, shell FEA, composite layup, structural checks

- Built an Abaqus shell model of a composite Formula SAE front wing using S4R elements, laminate definitions and local mesh refinement around attachment regions.
- Converted aero downforce/drag, lateral cornering and 4.1 G bump conditions into structural load cases for strength and stiffness checks.
- Checked ply-level stress utilisation against fatigue and ultimate allowables; the final laminate reached about 74% peak utilisation under the governing ultimate case with SF = 1.5.

Solar Chimney CFD Reproduction

ANSYS Fluent/CFX, conjugate heat transfer, natural convection

- Built a CFD reproduction workflow for a laboratory-scale solar chimney with coupled air, absorber and cover domains.
- Checked heat input, mesh sensitivity, mass balance, heat balance, Reynolds/Rayleigh scaling and solver behaviour before trusting the velocity results.
- Used the project to practise saying clearly what a model can support, what it cannot, and what physical data would be needed next.

Ruggedized Outdoor IoT Sensor Module

SolidWorks, enclosure design, DFM notes, Keil C, concept validation

- Designed a SolidWorks enclosure concept with screw bosses, gasket stack-up, PCB and battery placeholders, cable/strap mounts and service access.
- Added FEA-style screening for corner load, strap lug and screw boss regions; kept an open test list for sealing, impact, thermal exposure and firmware validation.
- Built concept firmware in Keil C with mock sensor readout, battery monitoring and a simple module state machine; documented the electronics and firmware architecture in the portfolio.

NZ Residential MVHR / ERV Ventilation Concept

SolidWorks, AutoCAD/DXF, HVAC sizing, NZBC G4/H1 context, commissioning plan

- Built a residential MVHR/ERV concept around a 3D duct layout, airflow schedule, duct velocity checks and preliminary pressure-loss estimate.
- Modelled the MVHR unit, supply/extract/outdoor/exhaust ducts, diffusers, grilles and hangers in SolidWorks; exported DXF, STEP, STL, PNG and a component list for review.
- Used 60 L/s balanced supply/extract as the normal case and an 85 L/s boost extract check; estimated about 80 Pa normal and 120 Pa boost external static pressure.
- Checked the design against NZBC G4/H1 and Healthy Homes ventilation context; added a commissioning test plan using CO2, humidity, airflow and fan-power measurements.

Orbital AI Data Centre Feasibility Study - Systems Lead

Stakeholders, requirements, ConOps, trade studies, risk, verification

- Led the systems-engineering side of a team concept for a 24-satellite LEO AI data-centre architecture.
- Set up the stakeholder map, requirements hierarchy, ConOps, architecture options, trade-study criteria, risk register and verification gates.
- Pulled subsystem work into one decision story across orbit/coverage, power, thermal rejection, communications capacity, launch mass, cost and sustainability.

TECHNICAL SKILLS

- Simulation: Abaqus/Standard, ANSYS Fluent, ANSYS CFX, shell FEA, CFD, conjugate heat transfer, natural convection
- Engineering checks: load cases, boundary conditions, mesh sensitivity, mass/heat balance, Reynolds/Rayleigh scaling, assumptions log
- CAD/design: SolidWorks, composite front-wing geometry, ventilation layouts, enclosure screening models
- Materials/structures: composite layup, ply utilisation, stiffness/strength checks, attachment-region refinement
- Communication: explaining what a model proves, what it does not prove, and what test data is needed next

LEADERSHIP AND COMMUNICATION

- Led team-level integration in the orbital data-centre project and commonly take the organiser role in group work.
- Former student news-department lead with photography/event documentation experience; comfortable turning technical work into clear visual evidence.
- Professional working proficiency in English; native Mandarin Chinese.